

Intro to deep learning

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Lecture 3: First experience with HuggingFace



What you will learn in this lecture

- How to use pre-trained models from huggingface
- How to clear the model cache to free up space
- How to set some tuning parameters to get what you want
- How to ask a model about the course logistics!

Ken Judd in Bonn

- OSE meetup, Thursday, 6 pm, no registration needed
- Office hours, Thursday afternoon, register [\[here\]](#)(IAME lounge (Lennestr. 43))
- Location of both: IAME lounge (Lennestr. 43)

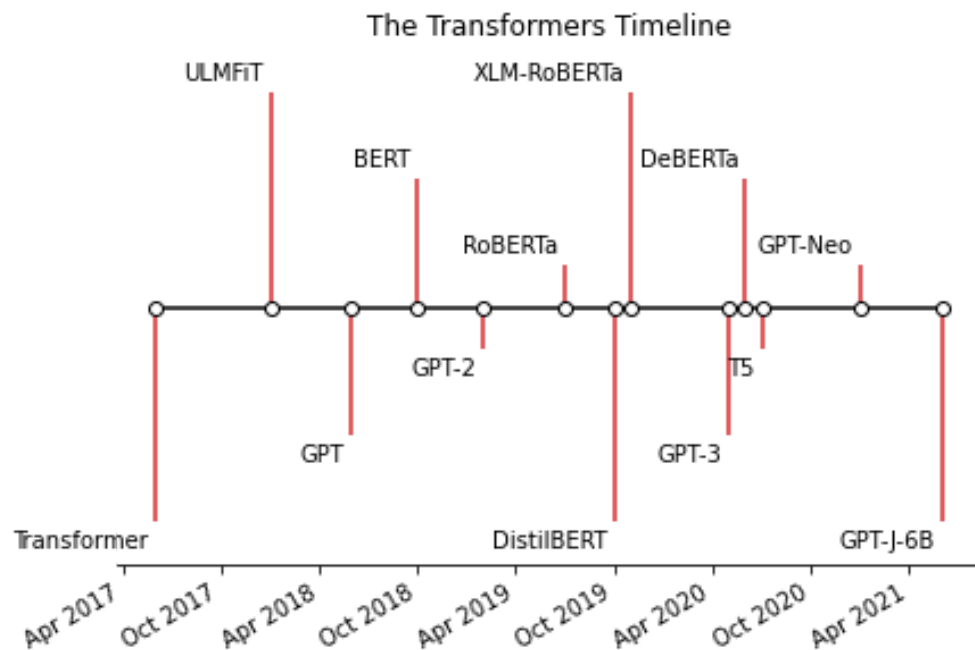
Should we start a bit earlier?

- Some students cannot attend second half
- Currently:
 - Start at 12:15
 - Break from 13:45 to 14:15
- Proposal
 - Start at 12:00
 - Have break from 14:00 to 14:30

A bit of history

- 2010: Start of deep learning revolution
 - CNNs win computer vision contests
 - Training on GPUs becomes mainstream
- Ca. 2015:
 - Transfer learning becomes mainstream in computer vision
 - DNNs outperform everything else in computer vision
- 2017 revolution starts in NLP:
 - Attention is all you need
 - ULMFiT brings transfer learning to NLP

Timeline



What is transfer learning

- Pre-training: Train a NN on giant amounts of **unlabeled data**
 - Language modelling: Predict next word in a sequence (e.g. GPT)
 - Masked language modelling: Predict a word that was removed (e.g. Bert)
- Domain adaptation: Continue on specialized text corpus
- Fine tuning:
 - Specialize last layer to the target task
 - Train it on **labeled data**

Plan for today

- Work with pre-trained and pre-fine-tuned models!
- Meet the most relevant NLP tasks:
 - Text classification / sentiment analysis
 - Named entity recognition
 - Question answering
 - Summarization
 - Translation
- Work with the transformers library for the first time

``transformers.pipeline``

- For us: Model is a black box that takes text and returns text
- In reality:
 - Text is converted to numbers (tokenizer)
 - Main model
 - Task specific head
 - Result is converted to text again
 - Post-processing
- ``transformers.pipeline`` hides all of this!

Important

- All the tasks ask you to use specific models instead of default
- This is to limit downloaded file-sizes

An example text

```
text = """Dear Amazon, last week I ordered an Optimus  
Prime action figure from your online store in Germany.  
Unfortunately, when I opened the package, I discovered  
to my horror that I had been sent an action figure of  
Megatron instead! As a lifelong enemy of the Decepticons,  
I hope you can understand my dilemma. To resolve the  
issue, I demand an exchange of Megatron for the Optimus  
Prime figure I ordered. Enclosed are copies of my records  
concerning this purchase. I expect to hear from you soon.  
Sincerely, Bumblebee."""
```

- Does the text have a negative or positive connotation?
- What people/objects are named in the text?
- What are main points of the text?
- What if we need a translation of the text?

Text classification

```
>>> from transformers import pipeline
>>> classifier = pipeline(task="text-classification")
>>> sentiments = classifier(text)
>>> sentiments
[{'label': 'NEGATIVE', 'score': 0.9015460014343262}]
```

- Create a `pipeline` with the task `"text-classification"`
- A task-specific default model is used, but you can define your model!
- The results are a lists of dictionaries
- Contain predictions and confidence scores.

Task 1

15 minutes

Named entity recognition

```
>>> ner_tagger = pipeline(task="ner")
>>> entities = ner_tagger(text)
>>> entities
[{'entity_group': 'ORG',
  'score': 0.9411546,
  'word': 'amazon',
  'start': 5,
  'end': 11},
 {'entity_group': 'MISC',
  'score': 0.9315696,
  'word': 'optimus prime',
  'start': 36,
  'end': 49},
  ...
]
```

- The task key is `"ner"`
- Additional arguments to define the pipeline
- The output is a list of dictionaries with recognized entities
- Entities are also classified as
 - Organizations
 - People
 - ...

Task 2

10 min

Optional arguments

```
>>> def add(x, y=2):  
...     return x + y
```

```
>>> add(x=4, y=5)  
9
```

```
>>> add(x=4)  
6
```

```
>>> add(y=4)
```

```
-----  
TypeError                                Traceback (most recent call la  
Cell In[15], line 4
```

```
1 def add(x, y=2):  
2     return x + y  
----> 4 add(y=4)
```

```
TypeError: add() missing 1 required positional argument: '
```

- Python functions can have optional arguments
- Use `=` sign in function signature to define them
- The default can be overwritten in function calls

Optional arguments in pipeline

- Which optional arguments can be used, depends on the model / task
- Sometimes a bit hard to find in the documentation
- For the NER task, we look at the argument ``aggregation_strategy``

(Default: `simple`). There are several aggregation strategies:

`none`: Every token gets classified without further aggregation.

`simple`: Entities are grouped according to the default schema (B-, I- tags get merged when the tag is similar).

`first`: Same as the `simple` strategy except words cannot end up with different tags. Words will use the tag of the first token when there is ambiguity.

`average`: Same as the `simple` strategy except words cannot end up with different tags. Scores are averaged across tokens and then the maximum label is applied.

`max`: Same as the `simple` strategy except words cannot end up with different tags. Word entity will be the token with the maximum score.

Pro-tip

- Always read the documentation of the libraries you are using
- Knowing and using optional arguments can set you apart from most others!
- Default values are not always chosen well
- Some arguments have a default value but should not

Task 3

(10 min)

Clearing the cache

- Models are large (200 MB to several GB) and are cached in `~/ .cache/huggingface``
- Even on your laptop, you will run out of space eventually
- The `hf`` command line tool lists and deletes cached models and datasets
- It is part of the course environment: run it with `uv run hf ...`` from the `dl_intro`` folder
- Same commands on all platforms

List cached models and datasets

```
uv run hf cache scan
```

- Prints one line per repository: id, type (model or dataset), size on disk, last use and local path
- `uv run hf cache scan -v`` additionally lists the cached revisions of each repository
- Set the environment variable `HF_HOME`` to move the cache to a different disk

Delete cached models and datasets

```
uv run hf cache delete
```

- Opens an interactive menu with all cached repositories
- Use the arrow keys to move, the space key to select and enter to continue
- Confirm the deletion by typing ``y``
- If the menu does not work in your terminal, use ``uv run hf cache delete --disable-tui``

Task 4

7 min

Question answering

```
>>> reader = pipeline(task="question-answering")
>>> question = "What does the customer want?"
>>> answers = reader(question=question, context=text)
>>> answers
{'score': 0.6312923431396484,
 'start': 335,
 'end': 358,
 'answer': 'an exchange of Megatron'}
```

- The task key is `"question-answering"`
- Define your question(s) as a string (or list of strings)
- Pass your question and the text as a context to the pipeline
- The output is a list of dictionaries with answers to questions

Task 5

12 min

Summarization

```
>>> summarizer = pipeline(task="summarization")
>>> summaries = summarizer(
...     text,
...     max_length=45,
...     min_length=45,
...     clean_up_tokenization_spaces=True
... )
>>> summaries
[{"summary_text": "Amazon's Optimus Prime action figure\ Wassent to a German online store in Germany. Bumblebee\ asks for an exchange of Megatron for the Optimus Prime\ figure he ordered. He expects to hear from you soon."}]
```

- The task key is `"summarization"`
- Summarize the text using additional arguments
- The output is a list of dictionaries with short summaries of the input text

Splitting and combining strings

```
zen = """The Zen of Python, by Tim Peters
```

```
Beautiful is better than ugly.  
Explicit is better than implicit.
```

```
Simple is better than complex.  
Complex is better than complicated."""
```

```
>>> parts = zen.split("\n\n")
```

```
>>> parts
```

```
['The Zen of Python, by Tim Peters',  
 'Beautiful is better than ugly.\nExplicit is better than  
 'Simple is better than complex.\nComplex is better than c
```

```
>>> print("\n---\n".join(parts))
```

```
The Zen of Python, by Tim Peters
```

```
---
```

```
Beautiful is better than ugly.  
Explicit is better than implicit.
```

```
---
```

```
Simple is better than complex.  
Complex is better than complicated.
```

- New lines are represented as ``\\n`` in python strings
- ``.split`` can split a string into a list of strings
- using ``"\n\n"`` as delimiter splits by paragraphs
- ``.join`` is the inverse of split

Task 7

(7 min)

https://huggingface.co/models

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Docs](#) [Solutions](#) [Pricing](#) [⌵](#) 

Tasks Libraries Datasets Languages Licenses Other

Multimodal

-  Feature Extraction  Text-to-Image  Image-to-Text
-  Text-to-Video  Visual Question Answering
-  Document Question Answering  Graph Machine Learning

Computer Vision

-  Depth Estimation  Image Classification
-  Object Detection  Image Segmentation
-  Image-to-Image  Unconditional Image Generation
-  Video Classification  Zero-Shot Image Classification

Natural Language Processing

-  Text Classification  Token Classification
-  Table Question Answering  Question Answering
-  Zero-Shot Classification  Translation
-  Summarization  Conversational
-  Text Generation  Text2Text Generation  Fill-Mask
-  Sentence Similarity

Audio

-  Text-to-Speech  Automatic Speech Recognition
-  Audio-to-Audio  Audio Classification
-  Voice Activity Detection

Tabular

-  Tabular Classification  Tabular Regression

Reinforcement Learning

-  Reinforcement Learning  Robotics

Models 183,501 [new](#) Full-text search [11 Sort: Most Downloads](#)

bert-base-uncased
 Updated Nov 16, 2022 ·  42.4M ·  744

 **Davlan/distilbert-base-multilingual-cased-ner-hrl**
 Updated Jun 27, 2022 ·  29.9M ·  40

xlm-roberta-base
 Updated 16 days ago ·  15.3M ·  253

 **microsoft/layoutlmv3-base**
 Updated 11 days ago ·  7.63M ·  131

roberta-base
 Updated Mar 6 ·  7.12M ·  152

 **openai/clip-vit-base-patch32**
 Updated Oct 4, 2022 ·  5.58M ·  166

xlm-roberta-large
 Updated 17 days ago ·  5.11M ·  131

bert-base-multilingual-cased
 Updated Nov 17, 2022 ·  3.96M ·  141

albert-base-v2
 Updated 5 days ago ·  2.94M ·  49

roberta-large
 Updated Mar 22 ·  2.27M ·  101

 **Helsinki-NLP/opus-mt-en-es**
 Updated Jan 24 ·  2.11M ·  37

 **google/electra-base-discriminator**
 Updated Apr 30, 2021 ·  2.05M ·  21

 **jonatasgrosman/wav2vec2-large-xlsr-53-english**
 Updated 29 days ago ·  40.4M ·  85

gpt2
 Updated Dec 16, 2022 ·  18.9M ·  928

 **openai/clip-vit-large-patch14**
 Updated Oct 4, 2022 ·  8.62M ·  347

distilbert-base-uncased
 Updated Nov 16, 2022 ·  7.37M ·  176

t5-base
 Updated 17 days ago ·  5.59M ·  198

distilroberta-base
 Updated Nov 17, 2022 ·  5.35M ·  62

bert-base-cased
 Updated Nov 16, 2022 ·  4.64M ·  95

 **prajjwal1/bert-tiny**
 Updated Oct 27, 2021 ·  3.14M ·  52

 **princeton-nlp/unsup-simcse-roberta-base**
 Updated Jun 16, 2021 ·  2.88M ·  8

 **runwayml/stable-diffusion-v1-5**
 Updated Jan 27 ·  2.17M ·  7.19k

 **cardiffnlp/twitter-roberta-base-sentiment**
 Updated Jan 20 ·  2.11M ·  158

 **pyannote/segmentation**
 Updated 24 days ago ·  1.91M ·  131

Use filters to find models

 **Hugging Face**

Models Datasets Spaces Docs Solutions Pricing

Tasks Libraries Datasets Languages Licenses Other

Reset Tasks

Multimodal

- Feature Extraction
- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
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- Unconditional Image Generation
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Natural Language Processing

- Text Classification
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- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Conversational
- Text Generation
- Text2Text Generation
- Fill-Mask
- Sentence Similarity

Models 4,372

new Full-text search Sort: Most Downloads

deepset/roberta-base-squad2

Updated 30 days ago · 1.31M · 272

distilbert-base-uncased-distilled-squad

Updated 17 days ago · 351k · 26

sultan/BioM-ELECTRA-Large-SQuAD2

Updated Aug 7, 2021 · 247k · 6

deepset/minilm-uncased-squad2

Updated Dec 5, 2022 · 106k · 29

deepset/roberta-base-squad2-covid

Updated Nov 18, 2022 · 87k · 5

monologg/koelectra-small-v2-distilled-korquad-384

Updated Jun 4, 2020 · 64.7k · 1

thatdramebaazguy/roberta-base-squad

Updated Jul 1, 2022 · 50.1k · 1

philschmid/distilbert-onnx

Updated Feb 16, 2023 · 30.4k · 1

bert-large-uncased-whole-word-masking-finetuned-squad

Updated 17 days ago · 612k · 81

distilbert-base-cased-distilled-squad

Updated 11 days ago · 304k · 112

deepset/bert-large-uncased-whole-word-masking-squad

Updated Dec 5, 2022 · 148k · 17

Rakib/roberta-base-on-cuad

Updated Jan 18 · 90.8k · 2

valhalla/longformer-base-4096-finetuned-squadv1

Updated Feb 10, 2021 · 86.1k · 10

deepset/roberta-large-squad2

Updated Jul 25, 2022 · 59.1k · 14

deepset/tinyroberta-squad2

Updated 30 days ago · 48.9k · 24

deepset/bert-base-cased-squad2

Updated Dec 5, 2022 · 30.4k · 1

Check model-card for more information

 **Hugging Face**

Search models, datasets, users...

Models Datasets Spaces Docs Solutions Pricing

deepset **roberta-base-squad2**  like 272

Question Answering PyTorch TensorFlow JAX Rust Safetensors Transformers squad_v2 English roberta Eval Results AutoTrain Compatible License: cc-by-4.0

Model card Files and versions Community 12

Train Deploy Use in Transformers

Edit model card

roberta-base for QA

This is the [roberta-base](#) model, fine-tuned using the [SQuAD2.0](#) dataset. It's been trained on question-answer pairs, including unanswerable questions, for the task of Question Answering.

Overview

Language model: roberta-base
Language: English
Downstream-task: Extractive QA
Training data: SQuAD 2.0
Eval data: SQuAD 2.0
Code: See [an example QA pipeline on Haystack](#)
Infrastructure: 4x Tesla v100

Downloads last month
1,309,088



Hosted inference API

Question Answering Examples

Where do I live? Compute

Context

My name is Wolfgang and I live in Berlin

Computation time on Intel Xeon 3rd Gen Scalable cpu: cached

Berlin 0.919

JSON Output Maximize

Dataset used to train deepset/roberta-base-squad2

squad_v2

Preview · Updated 18 days ago · ↓ 42.9k · ♥ 27

Spaces using deepset/roberta-base-squad2 194

microsoft/HuggingGPT seduerr/semantic_search

anakin87/who-killed-laura-palmer AmazonScience/QA-NLU

european-agency/play-with-transformer razakhan/text-summarizer

Hellisothepeople/HF-SHAP Eemansleedeprived/Study For Me AI

```
batch_size = 96
n_epochs = 2
base_LM_model = "roberta-base"
max_seq_len = 386
learning_rate = 3e-5
lr_schedule = LinearWarmup
warmup_proportion = 0.2
doc_stride=128
max_query_length=64
```

Translation

```
>>> translator = pipeline(task="translation_en_to_de")
>>> translations = translator(
...     text,
...     clean_up_tokenization_spaces=True,
...     min_length=50
... )
>>> translations
[{"translation_text": "Sehr geehrter Amazon,\nletzte Woche habe ich eine Optimus Prime Action\nFigur aus Ihrem Online-Shop in Deutschland bestellt.\nLeider, als ich das Paket öffnete, entdeckte ich zu\meinem Entsetzen, dass ich stattdessen eine Action\nFigur von Megatron geschickt worden war! Als\nlebenslanger Feind der Decepticons, Ich hoffe,\nSie können mein Dilemma verstehen. Um das Problem\nzu lösen, Ich fordere einen Austausch von Megatron\nfür die Optimus Prime Figur habe ich bestellt."}]
```

- The task key is `"translation_xx_to_yy"`
- Translate your text using additional arguments
- The output is a list of dictionaries with the translated text

Task 8

10 min